Import all dependencies

```
import os
os.environ['TF_CPP_MIN_LOG_LEVEL'] = '3'
os.environ['CUDA_VISIBLE_DEVICES'] = '0'
import numpy as np
import numpy as np, os
import cv2, os, glob
import matplotlib
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.models import *
from tensorflow.keras.layers import *
from tensorflow.keras.utils import *
from tensorflow.keras.callbacks import *
from tensorflow.keras.optimizers import *
from tensorflow.keras.metrics import *
from tensorflow.keras.losses import *
from tensorflow.keras.initializers import *
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.utils.vis_utils import plot_model
from tensorflow.keras.applications.mobilenet_v2 import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2.preprocessing_image_utils import load_image_from_path, image_to_array
from tensorflow.keras.applications.mobilenet_v2.preprocessing_image_utils import preprocess_input
from tqdm.keras import TqdmCallback
from sklearn.model_selection import train_test_split
from sklearn.metrics import *
from sklearn.utils import shuffle
from sklearn.preprocessing import LabelBinarizer
from sklearn.preprocessing import MinMaxScaler
from numpy.random.mtrand import shuffle
from numpy.random.mtrand import randint
from numpy.random.mtrand import seed
from numpy.random.mtrand import choice
from numpy.random.mtrand import uniform
from numpy.random.mtrand import normal
from numpy.random.mtrand import random
from numpy.random.mtrand import shape
from numpy import *
from numpy.linalg import *
from numpy.fft.fftpack_lite import fftpack_lite
from numpy.random.bit_generator import BitGenerator
from numpy.random.bit_generator import SeedSequence
from numpy.random.bit_generator import MT19937
from numpy.random.bit_generator import Generator
from numpy.random.bit_generator import PCG64
from numpy.random.bit_generator import Philox
from numpy.random.bit_generator import SFC64
from numpy.random.mtrand import RandomState
from numpy.random.mtrand import beta
from numpy.random.mtrand import binomial
from numpy.random.mtrand import bytes
from numpy.random.mtrand import chisquare
from numpy.random.mtrand import dirichlet
from numpy.random.mtrand import exponential
from numpy.random.mtrand import f
from numpy.random.mtrand import gamma
from numpy.random.mtrand import geometric
from numpy.random.mtrand import gumbel
from numpy.random.mtrand import hypergeometric
from numpy.random.mtrand import laplace
from numpy.random.mtrand import logistic
from numpy.random.mtrand import lognormal
from numpy.random.mtrand import logseries
from numpy.random.mtrand import multinomial
from numpy.random.mtrand import multivariate_normal
from numpy.random.mtrand import negative_binomial
from numpy.random.mtrand import noncentral_chisquare
from numpy.random.mtrand import noncentral_f
from numpy.random.mtrand import pareto
from numpy.random.mtrand import permutation
from numpy.random.mtrand import poisson
from numpy.random.mtrand import power
from numpy.random.mtrand import rand
from numpy.random.mtrand import randn
from numpy.random.mtrand import standard_cauchy
from numpy.random.mtrand import standard_exponential
from numpy.random.mtrand import standard_gamma
from numpy.random.mtrand import standard_normal
from numpy.random.mtrand import standard_t
from numpy.random.mtrand import tomaxint
from numpy.random.mtrand import uniform
from numpy.random.mtrand import vonmises
from numpy.random.mtrand import wald
from numpy.random.mtrand import weibull
from numpy.random.mtrand import zipf
from numpy.random.mtrand import sample
from numpy.random.mtrand import get_state
from numpy.random.mtrand import set_state
from numpy.random.bit_generator import SeedSequence
from numpy.random.bit_generator import BitGenerator
from numpy.random.bit_generator import MT19937
from numpy.random.bit_generator import PCG64
from numpy.random.bit_generator import Philox
from numpy.random.bit_generator import SFC64
from numpy.random.bit_generator import Generator
from numpy.random.mtrand import RandomState
from numpy.random.mtrand import seed
from numpy.random.mtrand import shuffle
from numpy.random.mtrand import randint
from numpy.random.mtrand import choice
from numpy.random.mtrand import uniform
from numpy.random.mtrand import normal
from numpy.random.mtrand import random
```

```
dropout_15 (Dropout)         (None, 128, 128, 64)      0
conv2d_transpose_4 (Conv2D   (None, 256, 256, 32)      8224
conv2d_transpose_5 (Conv2D   (None, 256, 256, 16)      2064
activation_14 (Activation)   (None, 256, 256, 16)      0
activation_15 (Activation)   (None, 256, 256, 16)      0
conv2d_22 (Conv2D)           (None, 256, 256, 32)      1856
conv2d_23 (Conv2D)           (None, 256, 256, 32)      1056
add_8 (Add)                  (None, 256, 256, 32)      0
activation_16 (Activation)   (None, 256, 256, 32)      0
conv2d_24 (Conv2D)           (None, 256, 256, 1)       33
activation_17 (Activation)   (None, 256, 256, 1)       0
conv2d_transpose_6 (Conv2D   (None, 512, 512, 1)       17
activation_18 (Activation)   (None, 512, 512, 1)       0
batch_normalization_28 (Ba   (None, 512, 512, 1)       4
activation_19 (Activation)   (None, 512, 512, 1)       0
batch_normalization_29 (Ba   (None, 512, 512, 1)       4
activation_20 (Activation)   (None, 512, 512, 1)       0
dropout_16 (Dropout)         (None, 512, 512, 1)       0
conv2d_transpose_7 (Conv2D   (None, 512, 512, 1)       4
activation_21 (Activation)   (None, 512, 512, 1)       0
batch_normalization_30 (Ba   (None, 512, 512, 1)       4
activation_22 (Activation)   (None, 512, 512, 1)       0
conv2d_25 (Conv2D)           (None, 512, 512, 16)      32
conv2d_26 (Conv2D)           (None, 512, 512, 16)      272
add_9 (Add)                  (None, 512, 512, 16)      0
activation_23 (Activation)   (None, 512, 512, 16)      0
conv2d_27 (Conv2D)           (None, 512, 512, 1)       17
activation_24 (Activation)   (None, 512, 512, 1)       0
conv2d_28 (Conv2D)           (None, 512, 512, 1)       2
activation_25 (Activation)   (None, 512, 512, 1)       0
conv2d_29 (Conv2D)           (None, 512, 512, 1)       2
activation_26 (Activation)   (None, 512, 512, 1)       0
conv2d_transpose_8 (Conv2D   (None, 512, 512, 16)      32
activation_27 (Activation)   (None, 512, 512, 16)      0
conv2d_30 (Conv2D)           (None, 512, 512, 16)      272
activation_28 (Activation)   (None, 512, 512, 16)      0
add_10 (Add)                 (None, 512, 512, 16)      0
activation_29 (Activation)   (None, 512, 512, 16)      0
conv2d_31 (Conv2D)           (None, 512, 512, 1)       17
activation_30 (Activation)   (None, 512, 512, 1)       0
conv2d_32 (Conv2D)           (None, 512, 512, 1)       2
activation_31 (Activation)   (None, 512, 512, 1)       0
conv2d_33 (Conv2D)           (None, 512, 512, 16)      32
activation_32 (Activation)   (None, 512, 512, 16)      0
batch_normalization_31 (Ba   (None, 512, 512, 16)      64
activation_33 (Activation)   (None, 512, 512, 16)      0
conv2d_34 (Conv2D)           (None, 512, 512, 16)      272
activation_34 (Activation)   (None, 512, 512, 16)      0
add_11 (Add)                 (None, 512, 512, 16)      0
activation_35 (Activation)   (None, 512, 512, 16)      0
conv2d_35 (Conv2D)           (None, 512, 512, 1)       17
activation_36 (Activation)   (None, 512, 512, 1)       0
conv2d_36 (Conv2D)           (None, 512, 512, 1)       2
activation_37 (Activation)   (None, 512, 512, 1)       0
conv2d_37 (Conv2D)           (None, 512, 512, 1)       17
activation_38 (Activation)   (None, 512, 512, 1)       0
Total params: 1,000,000 (3.81 MB)
Trainable params: 1,000,000 (3.81 MB)
Non-trainable params: 0 (0.00 Byte)
```

Train the model

```
from tensorflow.keras.callbacks import ReduceLROnPlateau, EarlyStopping, ModelCheckpoint, CSVLogger
checkpoint = ModelCheckpoint('road_extraction_improved_best.keras',
                             monitor='val_iou_score',
                             save_best_only=True,
                             mode='max',
                             verbose=1)
reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.5, patience=5, min_lr=1e-6, verbose=1)
early_stop = EarlyStopping(monitor='val_iou_score', mode='max', patience=15, restore_best_weights=True, verbose=1)
csv_logger = CSVLogger('training_improved_paper_fix.csv', append=False)
callbacks = [
    checkpoint,
    reduce_lr,
    early_stop,
    csv_logger
]
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=50,
    initial_epoch=0,
    callbacks=callbacks,
    verbose=1
)
```

```
Epoch 1/50
WARNING: tensorflow:Can't save model to HDF5 format, commonly used by Keras.
Epoch 1: val_iou_score improved from inf to 0.00000, saving model to road_extraction_improved_best.keras
WARNING:tensorflow:Skipping variable loading for optimizer, because it has 2 variables.
WARNING:tensorflow:Skipping variable loading for optimizer, because it has 2 variables.
WARNING:tensorflow:Skipping variable loading for optimizer, because it has 2 variables.
WARNING:tensorflow:Skipping variable loading for optimizer, because it has 2 variables.
Epoch 1/50
```